

Comparison technique for semilinear problems



Fernán González-Ibáñez joint work with José María Espinar and Diego A. Marín

Universidad de Granada

Research partially supported by MCIU/AEI/10.13039/501100011030, the FSE+ and Spanish MIC Grant PID2023-150727NB-I00, María de Maeztu Excellence Unit IMAG, reference CEX2020-001105-M, funded by MCINN/AEI/10.13039/501100011030/CEX2020-001105-M, and Spanish MIC Grant PID2024-160586NB-I00.

The Set Up

- Let (M^n, g) be a Riemannian manifold such that for $k \in \mathbb{R}$,
$$\text{Ric}_g \geq k(n-1)g.$$
- Let $\Omega \subset M^n$ be a paracompact domain with C^2 boundary.
- Let $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ be a locally Lipschitz function such that

$$f(x) > 0 \quad \text{and} \quad f(x) \geq nkx + f(0) \quad \text{for } x \in I_f,$$

with $f(0) > 0$ when $k \leq 0$.

Then we consider the following **semilinear elliptic problem**,

$$\begin{cases} \Delta u + f(u) = 0 & \text{in } \Omega, \\ u > 0 & \text{in } \Omega, \\ u = 0 & \text{in } \partial\Omega. \end{cases} \quad (\star)$$

Basis for the Comparison

Model manifold

$$(\mathcal{M}_k^n(F), g_k) := (I_k \times F, dr^2 + s_k(r)^2 g_F).$$

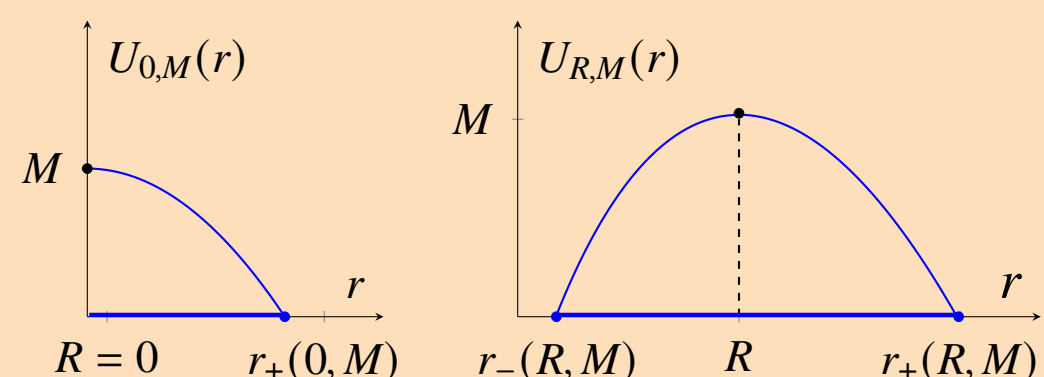
The Laplace–Beltrami operator on a model manifold,

$$\Delta = \partial_r^2 + (n-1) \cot_k(r) \partial_r + \frac{1}{s_k(r)^2} \Delta_F.$$

Model Solutions (*Radial symmetric solutions*)

$$u(p) = U(r(p)) \quad \forall p \in \Omega \Leftrightarrow \begin{cases} U''(r) + (n-1) \cot_k(r) U'(r) + f(U(r)) = 0, \\ U(R) = M, \quad U'(R) = 0. \end{cases} \quad (\star)$$

For each initial data (R, M) , we have two types of solutions, called **model pairs** denote by Model_M ,



and two kinds of domains:

$$\Omega_{R,M} := \begin{cases} \{p \in \mathcal{M}_k^n(F) : r(p) < r_+(0, M)\} & \text{if } R = 0 \\ \{p \in \mathcal{M}_k^n(F) : r_-(R, M) < r(p) < r_+(R, M)\} & \text{if } R > 0. \end{cases}$$

Then, $\mathfrak{U}_{R,M}^\pm \subset \Omega_{R,M} \setminus \Gamma_{R,M}$ the subdomains of $\Omega_{R,M}$ without maximum points and the corresponding solutions of $(\star)$ restricted to $\mathfrak{U}_{R,M}^\pm$ is denoted as $u_{R,M}^\pm(p)$.

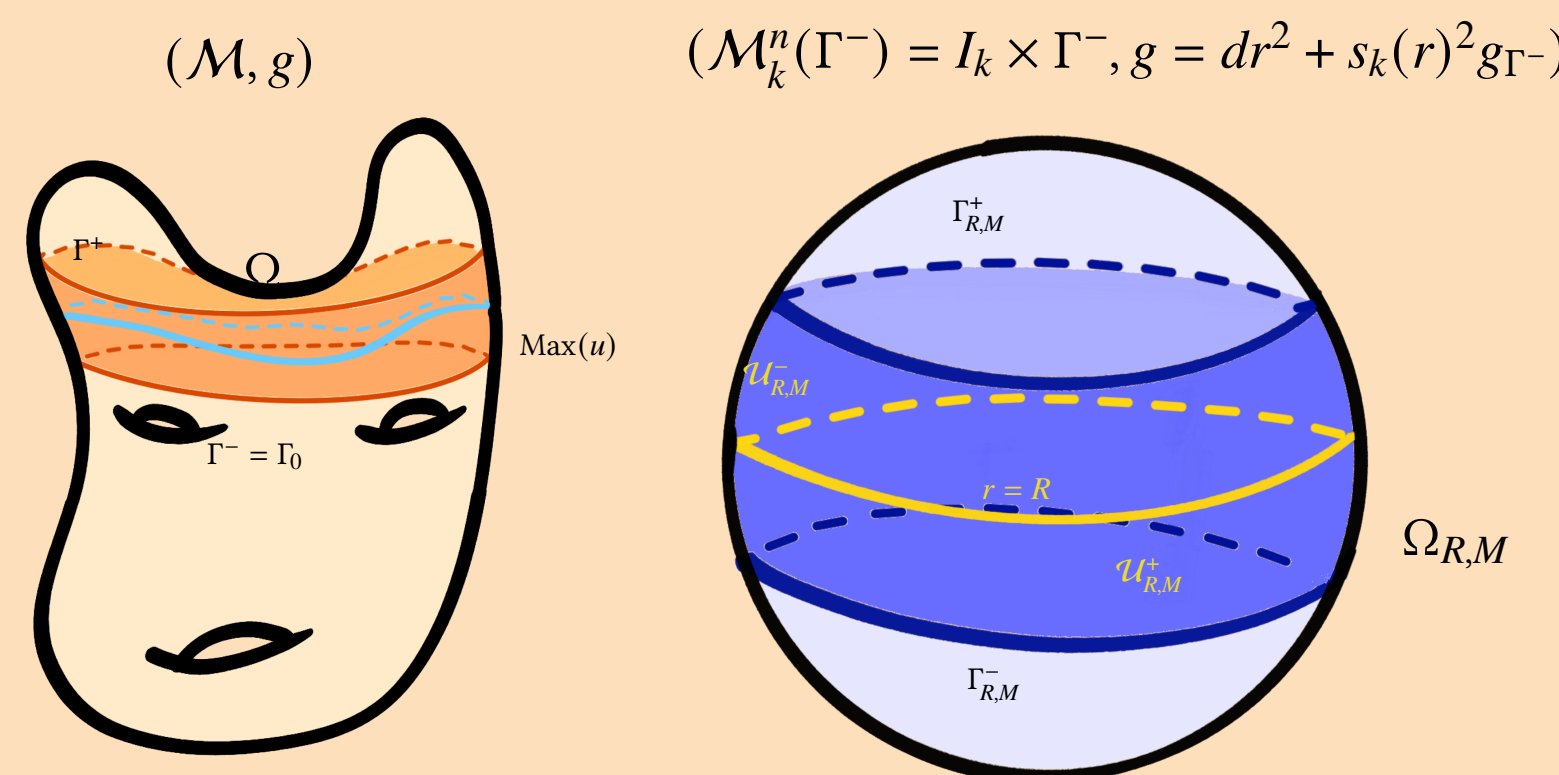
Model $\bar{\tau}$ -functions are defined as,

$$\bar{\tau}_M^\pm(R) := \frac{U_{R,M}'(r_\pm(R, M))^2}{U_{0,M}'(r_+(0, M))^2}$$

Comparison algorithm

We fix $\mathfrak{U}$ be a connected component of $\Omega \setminus \text{Max}(u)$. For each connected component of the boundary $\Gamma \in \pi_0(\partial\Omega)$, define

$$\bar{\tau}(\Gamma) := \max_{p \in \Gamma} \frac{|\nabla u(p)|^2}{U_{0,M}''(r_+(0, M))^2}.$$



We say that a model pair in $\mathcal{M}_k^n(\Gamma_0)$ $(\tilde{\mathfrak{U}}, \tilde{u}) := (\mathfrak{U}_{R,M}^\circ, u_{R,M}^\circ) \in \text{Model}_M^\circ$, where

$\diamond \in \{-, +\}$ is a,

► a *comparison pair* for $(\mathfrak{U}, u)$ if $\bar{\tau}(\mathfrak{U}) \leq \bar{\tau}(\tilde{\mathfrak{U}}) = \bar{\tau}_M^\diamond(R)$,

► An *associated model pair* if $\bar{\tau}(\mathfrak{U}) = \bar{\tau}(\tilde{\mathfrak{U}}) = \bar{\tau}_M^\diamond(R)$

The comparison tool

- We fix $(\tilde{\mathfrak{U}}, \tilde{u})$ a comparison pair to $(\mathfrak{U}, u)$ with core radius $\bar{R}$.

We define the function,

$$G : [0, M] \times [\bar{r}_-, \bar{r}_+] \rightarrow \mathbb{R} \\ (s, r) \rightarrow G(s, r) := s - \bar{U}(r).$$

By the Implicit Function Theorem, there exist two C^2 -functions,

$$\chi^- : [0, M] \rightarrow [\bar{r}_-, \bar{R}] \quad \chi^+ : [0, M] \rightarrow [\bar{R}, \bar{r}_+]$$

$\chi^\pm$ is the inverse of $\bar{U}$ on the corresponding monotonicity interval.

The *pseudo-radial function* associated to $(\mathfrak{U}, u)$ and $(\tilde{\mathfrak{U}}, \tilde{u})$ is

$$\Psi : \mathfrak{U} \rightarrow \chi^\circ(0, M) \subset [\bar{r}_-, \bar{r}_+], \quad \Psi(p) := \chi^\circ(u(p)).$$

Notation

Let $I_k := [0, \bar{r}_k)$, where $\bar{r}_k := +\infty$ if $k \leq 0$ and $\bar{r}_k := \pi/\sqrt{k}$ if $k > 0$.

$$s_k(r) := \begin{cases} \frac{\sinh(\sqrt{-k}r)}{\sqrt{-k}} & \text{if } k < 0 \\ r & \text{if } k = 0, \\ \frac{\sin(\sqrt{k}r)}{\sqrt{k}} & \text{if } k > 0, \end{cases} \quad \cot_k(r) := \frac{s_k'(r)}{s_k(r)} \quad \text{and} \quad \tan_k(r) = \frac{s_k(r)}{s_k'(r)}.$$

Comparison results

Gradient estimates

Let $f \in C^1(\mathbb{R})$ be a function satisfying condition:

$$f'(x) \geq nk \quad \text{for all } x \in I_f. \quad ()$$

We have

$$|\nabla u|^2(p) \leq \bar{U}'(\Psi)^2(p) \quad \text{for all } p \in \mathfrak{U}.$$

If equality holds at any point $p \in \mathfrak{U}$, $\begin{cases} 1. \text{Ric}(\nabla u, \nabla u) = (n-1)k |\nabla u|^2 \text{ on } \mathfrak{U}, \\ 2. (\mathcal{M}^n, g) \text{ is isometric to } (\mathcal{M}_k^n(\Gamma_0), g_k) \text{ on } \mathfrak{U}, \\ 3. (\mathfrak{U}, u) \text{ is congruent to } (\tilde{\mathfrak{U}}, \tilde{u}). \end{cases}$

Location of hot-spots

Let $f \in C^1(\mathbb{R})$ the condition $()$ and

$$f(x) \leq f_k(x) = nkx + 1$$

Assume that Ω is contained in a convex domain of $\mathcal{M}$ and that $\bar{\tau}(\Omega) = \bar{\tau}(\mathfrak{U})$. If $k > 0$ suppose that $\bar{\tau}_+ \leq \bar{r}_k/2$. Then,

$$\frac{\text{dist}(p, \partial\Omega)}{\tan_k(r_\Omega)/n} \begin{cases} \geq \frac{\bar{r}_+ - \bar{R}}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (U, \bar{u}) \in \bar{\text{Model}}_M^+, \\ > \frac{\bar{R} - \bar{r}_-}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (U, \bar{u}) \in \bar{\text{Model}}_M^-, \end{cases} \quad \text{for all } p \in \text{Max}(u).$$

where $r_\Omega = \max \{\text{dist}(p, \partial\Omega) : p \in \Omega\}$. If equality holds at a point of Ω , then:

- $(\mathcal{M}, g)$ is isometric to the space form $\mathbb{M}^n(k)$,
- $f(x) = nkx + 1$,
- (Ω, u) is a ball solution of Serrin's problem.

Isoperimetric type inequality

Let $f \in C^m(\mathbb{R})$, with $m = \max\{n-2, 3\}$, satisfying condition $()$. Assume that the $(n-1)$ -dimensional part of $\text{cl}(\mathfrak{U}) \cap \text{Max}(u)$, denoted by Γ_M , consists of a possibly disconnected Lipschitz-continuous hypersurface. Hence,

$$\frac{\mathcal{H}^n(\mathfrak{U})}{\mathcal{H}^{n-1}(\Gamma_M)} \geq \begin{cases} \frac{\mathcal{H}_k^n(\mathfrak{U}_{R,M}^+)}{\mathcal{H}_k^{n-1}(\Gamma_{R,M}^+)} & \text{if } \tilde{\mathfrak{U}} \in \text{Model}_M^+, \\ \frac{\mathcal{H}_k^n(\mathfrak{U}_{R,M}^-)}{\mathcal{H}_k^{n-1}(\Gamma_{R,M}^-)} & \text{if } \tilde{\mathfrak{U}} \in \text{Model}_M^-. \end{cases}$$

Equality holds if and only if $(\mathfrak{U}, u) \equiv (\tilde{\mathfrak{U}}, \tilde{u})$.

Bibliography

- V. Agostiniani, S. Borghini, and L. Mazziere. *On the Serrin problem for ring-shaped domains*.
- J.M. Espinar, F. G.-I. and D.A. Marín *Comparison methods for semilinear elliptic problems on Riemannian manifolds with a Ricci lower bound*.
- R. Magnanini and G. Poggesi, *The location of hot spots and other extremal points*.